HugIpuNet: Hybrid Updating Graph Motivated Illumination-Wise Property Unification Network for Hyperspectral and DSM Joint Classification

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***Abstract*—Hyperspectral images (HSIs) and digital surface models (DSMs) derived from light detection and ranging (LiDAR) data are packed with rich on-ground object characteristics, making their joint classification crucial for accurate remote sensing (RS) identification. However, variations in illumination properties hinder efficient and precise cross-modal information interaction. This phenomenon further exacerbates the difficulty in correctly identifying corresponding spectral curves, ultimately resulting in suboptimal performance. In this article, a hybrid updating graph motivated illumination-wise property unification network (HugIpuNet) is proposed to solve these challenges. Our core motivation is to unify global illumination descriptions with assistance from elevations for capturing precise multimodal properties from HSIs and DSMs derived from LiDAR data. HS spectral curves and LiDAR elevations interact constantly, considering their geological connection to strengthen their coupling capability. HugIpuNet further considers several illumination-wise mechanisms to address intricate and variable environmental conditions. The whole structure utilizes various hybrid updating (HU) graphs to enhance the attribute unification ability in complex environmental scenarios. Dynamic adjacent matrices are constructed to introduce uncertainty to limit the effect caused by description distortions. Experiments show an average 2.86% improvement in overall accuracy (OA) compared with multiple state-of-the-art approaches. The detailed datasets and codes will be available at https://github.com/weixinttt/HugIpuNet**

***Index Terms*—Classification, environmental illumination, hybrid graph updating, hyperspectral, light detection and ranging (LiDAR).**

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I. Introduction

REMOTE sensing (RS) has increasingly become an indispensable component in diverse fields, facilitating information and gathering comprehensive landscape analysis through multiple spaceborne/airborne probing tools. Diverse RS modalities, including hyperspectral images (HSIs) [1] and digital surface models (DSMs) derived from light detection and ranging (LiDAR) data [2], can be captured by RS techniques. Notably, HSIs with spatial and spectral characteristics are optical information sources for land-cover identification. Their capabilities can be affected by environmental illumination, considering their imaging modes. Meanwhile, LiDAR data are obtained by actively emitting pulsed laser light and measuring its reflection time from objects, thereby capturing detailed elevations and 3-D coordinates. Thereby, DSMs are utilized to describe terrain and artificial surfaces for lightweighting. Previous research proves that the integration of HSIs and DSMs can leverage elevations to improve the identification capability. The obtained multimodal modality contributes to capturing critical information and mitigating issues associated with external factors.

Deep learning (DL)-based techniques have become pivotal tools in joint classification tasks [3], [4]. Their fusion accuracies can be influenced by several factors, for example, illumination angle and imaging time. Some works tend to use diverse generative models [5] to enhance fusion robustness and minimize performance fluctuation. Chen et al. [6] first introduced a stage-wise diffusion model framework to perceive and utilize spatial–spectral information. Considering redundant HS bands always make extracted features lack specificity, convolutional neural networks (CNNs) play crucial roles in the related works [7]. Xu et al. [8] proposed a cross-attention fusion network by bridging CNN and transformer, further exploring both local and global cross-modal information. The long-distance feature information for accurate and efficient key feature learning is valuable for consideration. In contrast, generative models focused on the reconstruction procedure rather than directly obtaining classification accuracy by simulating the relative RS modality dynamic distribution and classified categories. Specifically, an autoencoder (AE) [9], [10], [11] is indicated as adopting underlying distributions and setting the corresponding constraints in the encoder to limit the latent distribution. Inspired by these works, transformer

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[12], [13], [14], [15] demonstrates superior performance for large-scale RS modalities, effectively and efficiently extracting global descriptions. To fully leverage their advantages in HSIs, Zou et al. [16] presented a novel pyramid superpixel transformer for classification. The authors verified that, as the feature attributes become increasingly complex, the global information capturing ability becomes particularly significant.

The attention-related mechanisms in the transformer always compute the interactions among individual elements within a single sequence, leading to excessive memory and time consumption. The appearance of the state-space model (SSM)-based Mamba has significantly enhanced computational efficiency by utilizing sub-quadratic time structures. This innovative structure transforms SSM parameters into input functions, thereby optimizing performance. Li et al. [17] presented one Mamba-based HS classification network (MambaHSI). They introduced the spatial Mamba block (SpaMB) and spectral Mamba block (SpeMB) for adaptively integrating information, enabling more flexible data processing steps. Furthermore, He et al. [18] presented a lightweight interval group spatial–spectral Mamba framework to extract global information in a grouping and hierarchical manner. Apparently, these techniques offer great flexibility in identifying key points within complex structures and emphasize that both global and local information are indispensable in joint classification tasks. To address the limitation of incomplete global features, some works also consider graph convolutional networks (GCNs) since they establish connections through adjacency matrices effectively in irregularly shaped regions [19], [20], [21], [22].

Existing multimodal fusion techniques have achieved notable progress. However, two primary challenges, that is, illumination divergence and heterogeneous association, still hinder exploring high-quality RS modality fusion performance.

On the one hand, the uncertainty associated with HS illumination properties always leads to severe spectral characteristic distortion, ultimately resulting in poor classification performance. Furthermore, illumination occlusion issues together with land-cover variation always complicate global–local information acquisition in joint fusion and classification tasks. On the other hand, joint classification requires exchanging and integrating numerous latent features to guarantee classification accuracy and modality consistency. However, HSIs and DSMs are captured by diverse RS instruments and possess distinct structural characteristics. Fusing them directly by existing approaches cannot attain satisfactory performance. A hybrid updating graph motivated illumination-wise property unification network (HugIpuNet) is presented to address these challenges. Its motivation is to unify global illumination descriptions with assistance from LiDAR elevations to capture precise multimodal properties from HSIs and DSMs. Notably, HugIpuNet consists of four modules: the characteristic pattern simulation (CPS) module, the hybrid updating graph convolution network (HUGCN) module, the spectrum distribution localization (SDL) module, and the decomposition linear combination (DLC) module. To address the illumination divergence problem, HugIpuNet employs a comprehensive approach utilizing spatial–spectral elevation information to explore the correlation between corresponding HS spectral curves and LiDAR elevations, as shown in Fig. 1. This strategy aims to mitigate the effect of illumination properties on land-cover interpretation. Meanwhile, the geological connection between these two RS modalities is fully considered to strengthen their coupling capability for satisfactory fusion performance. The hybrid updating (HU) GCN-related structures in both the HUGCN and SDL modules contribute to enhancing the attribute unification ability in complex environmental scenarios. The SDL module further considers SSM-based structures and dynamic adjacent matrices for long-range dependency enhancement and global information analysis. The DLC module successfully decomposes HSIs into reflectance and shading components with the help of DSMs derived from LiDAR data, further increasing the fusion interpretability of HugIpuNet. These illumination-wise mechanisms perform well on precise cross-modal information interaction in real-world tasks. The main contributions are listed as follows.

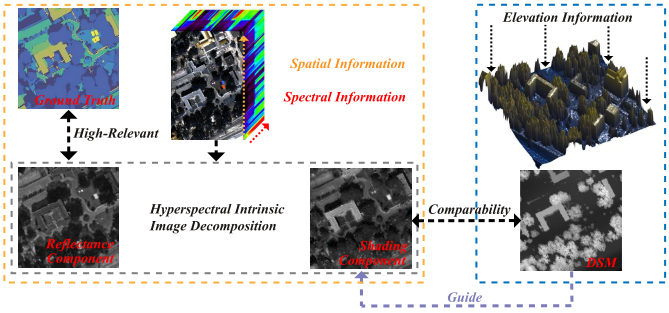


Fig. 1. HSI and DSM information complementarity in classification tasks.

1) To strengthen the coupling capability between both RS modalities and unify global illumination descriptions in complex environmental scenarios, one novel network called HugIpuNet for HS and DSM joint classification is proposed. To the best of our knowledge, this article is the first work that combines highly adaptive illumination description strategies and dynamic HU graph construction mechanisms to enhance the cross-modal information interaction ability in real-world tasks.
2) HugIpuNet precisely limits the effects caused by on-ground object description distortions in complex environmental scenarios and generates effective global–local information to dynamically enhance HS and DSM sample correlations in both the HUGCN and SDL modules by generating specific fluctuating adjacent matrices and constructing HU graphs. In addition, to enhance the robustness of HugIpuNet, an uncertainty mechanism is adopted in the adjacent matrix construction process.
3) HugIpuNet emphasizes the geological connection to generate high-quality reflectance and shading components between HSIs and DSMs derived from LiDAR data in complex environmental scenarios. The decomposed shading components are further concatenated with other extracted features to strengthen object arrangement information and avoid information loss. The

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decomposed reflectance components mainly ensure joint classification performance. Multiple specific constraints are incorporated in the loss function to guide the training direction of HugIpuNet.

Section II provides some related works. Section III describes HugIpuNet in detail. Section IV provides and analyzes the experimental results. Section V concludes this work.

II. Related Works

A. HS and DSM Joint Classification

Several studies have shown that fusing HSIs and DSMs derived from LiDAR data is beneficial for accurately identifying on-ground object categories in real-world tasks. Existing works attempt to enhance the RS modality connection for satisfactory joint classification performance [23]. Guo et al. [24] embedded HSIs and DSMs in the same feature space, aligning multisource distributions to alleviate data description differences. Liao et al. [25] captured semantic features from several RS modalities and edge contour data by Mamba-based networks. Moreover, some long-distance information interaction guided works also achieve superior results [26], [27]. Nonetheless, the modality heterogeneity between HSIs and DSMs hinders capturing comprehensive and precise discriminant features, making it hard to leverage the multimodal advantages. In this article, HugIpuNet reduces the impact of this modality heterogeneity on classification performance by enhancing the geological connection between these two RS modalities and strengthening their coupling capability.

B. HS Intrinsic Image Decomposition

HS intrinsic image decomposition (HIID) decomposes one HSI into one reflectance component $\boldsymbol{R}$ and one shading component $\boldsymbol{S}$ to capture environmental illumination descriptions. Studies have shown that HIID [28], [29], [30] is indispensable for analyzing spectral characteristics, especially when using large-scale RS optical images with inevitable interference. For one HSI $\boldsymbol{H} \in \mathbb{R}^{a\times b\times c}$, where a, b, and c are the height, width, and HS channel number, HIID can be formulated as

$$\boldsymbol{H}(\lambda) = \boldsymbol{R}(\lambda) \odot \boldsymbol{S}(\lambda), \quad 1 \le \lambda \le c \tag{1}$$

where $\odot$ is the Hadamard product, and λ stands for the channel index corresponding to wavelength. The decomposed $\boldsymbol{R}$ and $\boldsymbol{S}$ reflect the material differences in uniform illumination and the specific environmental illumination descriptions, respectively. Gong et al. [31] presented a deep HIID with adversarial learning for HS classification by mitigating environmental factors. In this article, HugIpuNet retains $\boldsymbol{S}$ to unify illumination-wise properties between HSIs and DSMs and strengthens their geometric connection. This is motivated by the fact that nonuniform illumination inherently describes object arrangement information for accurate identification to some extent.

C. GCN

GCN is excellent at extracting global characteristics in complex-scale classification tasks. It has strong adaptive capacities when dealing with object contours. Moreover, GCN accurately captures key characteristics since it requires the participation of the entire graph for a global vision within the graph construction procedure. By constructing nodes and edges, GCN ensures that its adjacency matrix is directly related to its graph structure. Additionally, GCN enhances its interpretability through its specific optimization capability, which is based on the structure of surrounding regions. Huang et al. [32] found that GCN works well in exploiting neighborhood vertex interactions based on its unique aggregation mechanism. Lu et al. [33] attempted to propose a separable deep GCN integrated CNN to extract global structural information. Feng et al. [34] designed a multichannel GCN (MGCN) for batch-wise network training. In this article, HugIpuNet considers GCN for capturing abundant spectral information from HSIs and DSMs and reflecting identification functions. However, these two heterogeneous RS modalities contain high-level complex spatial–spectral elevation information, imposing high requirements for information alignment.

D. SSM and Mamba

SSMs have essential advantages when dealing with long-distance dependency issues. Considering HSIs contain multiple spectral channels, SSM is capable of reducing computational complexities and improving parallel computing efficiencies. Specifically, by incorporating forgetting gates and blocking mechanisms, SSM-based Mamba structures have the capability of deviation detection and location recognition, excellently improving their inference speeds and parallel computation capabilities. Liu et al. [35] presented a novel HyperMamba structure to adaptively scan spatial neighborhood pixels and dynamically enhance spectral channels for spectral scanning. Yang et al. [36] proposed a GraphMamba processing paradigm to preserve spatial–spectral features by constructing cubes. In this article, HugIpuNet fully utilizes the adaptive scanning capability of Mamba and designs specific scanning mechanisms for both RS modalities, considering their differences. Furthermore, in the spectral perspective, GCN with strong neighborhood utilizing capability is adaptive to Mamba structures in RS modality fusion tasks, making fused information abundant and reasonable. Thus, HugIpuNet attempts to connect GCN with Mamba to enhance the modality fusion interpretability and yield superior performance.

III. Methodology

The scheme of HugIpuNet with detailed structures and annotations is provided in Fig. 2. In general, HugIpuNet consists of four modules, including CPS, HUGCN, SDL, and DLC. HSI and one typical lightweight LiDAR modality, that is, DSM, are selected for joint classification.

A. CPS Module

The CPS module aims to capture high-quality information from the HSI and DSM for the following modules. Formally, the HSI and LiDAR-derived DSM are denoted as $\boldsymbol{H} \in \mathbb{R}^{a\times b\times c}$ and $\boldsymbol{L} \in \mathbb{R}^{a\times b}$, respectively, where a, b, and c are the length, width, and HS channel number. To extract sufficient spatial features within neighboring regions, both $\boldsymbol{H}$ and $\boldsymbol{L}$

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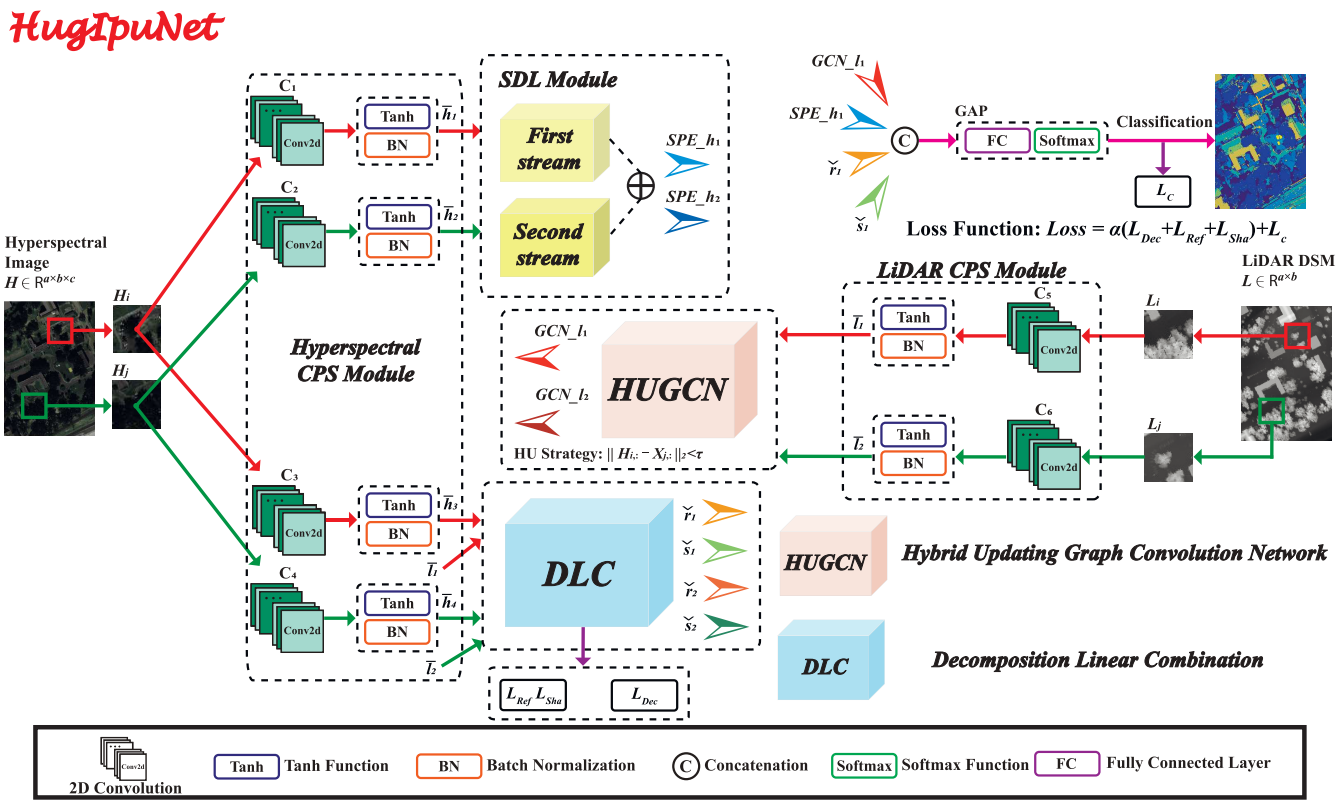


Fig. 2. Scheme of HugIpuNet for HS and DSM joint classification. The whole network is composed of four modules, including CPS, HUGCN, SDL, and DLC. The layers related to these two patch pairs are marked by red and green lines, respectively. The classification-related layers are marked by pink lines.

are partitioned into several $w \times w$ patches, where w is the sliding window side. The ith patch pair in both $\boldsymbol{H}$ and $\boldsymbol{L}$ can be represented as $\boldsymbol{H}_i \in \mathbb{R}^{w\times w\times c}$ and $\boldsymbol{L}_i \in \mathbb{R}^{w\times w}$, where $1 \le i \le a \times b$. Considering that HugIpuNet attempts to measure local and global gradients, the ith and jth patch pairs are randomly selected in the CPS module ($1 \le i, j \le a \times b$ and $i \ne j$). This module contains two streams. 1) The HS-related stream consists of four 2-D convolution (Conv2d) layers, that is, C_1, C_2, C_3, and C_4. Notably, C_1 and C_2 are for spectrum distribution localization while C_3 and C_4 are for cross-modal illumination description. Capturing spectral and spatial features for fusion is beneficial in obtaining sufficient and comprehensive discriminant information. The inputs for C_1 and C_3 are $\boldsymbol{H}_i$. The inputs for C_2 and C_4 are $\boldsymbol{H}_j$. The Conv2d layers in this article always contain $c3 \times 3$ kernels (SAME padding). 2) The LiDAR-related stream consists of two Conv2d layers, that is, C_5 and C_6, for capturing elevation information from $\boldsymbol{L}_i$ and $\boldsymbol{L}_j$. Their inputs are $\boldsymbol{L}_i$ and $\boldsymbol{L}_j$. The outputs of all Conv2d layers, $\bar{\boldsymbol{h}}_1$, $\bar{\boldsymbol{h}}_2$, $\bar{\boldsymbol{h}}_3$, $\bar{\boldsymbol{h}}_4$, $\bar{\boldsymbol{l}}_1$, and $\bar{\boldsymbol{l}}_2$, are activated by the Tanh function and normalized by batch normalization. The output shapes of the CPS module are $[\mathrm{BS}, w, w, c]$, where BS is the batch size.

B. HUGCN Module

Complex environmental scenarios always interfere with extracting effective global–local information and obscuring some relevant edge features. Fig. 3 is the core architecture of the HUGCN module, which attempts to tackle these challenges and enhance DSM sample correlations dynamically by generating specific adjacent matrices and constructing HU graphs adaptively. The obtained $\bar{\boldsymbol{l}}_1$ and $\bar{\boldsymbol{l}}_2$ are processed by two Conv2d layers C_{11} and C_{12} initially and are delivered into the HUGCN module with two inputs and two outputs. In particular, the structures for $\bar{\boldsymbol{l}}_1$ and $\bar{\boldsymbol{l}}_2$ share the same HUGCN parameters. These inputs are flattened to two $[\mathrm{BS}, w^2, c]$ vectors. This module consists of two stacked HUGCN layers in a single stream, which strengthens regional feature relationships by generating HU node and edge sets. For a clear description, the input of the HUGCN layer is denoted as $\boldsymbol{X} \in \mathbb{R}^{w^2\times c}$ directly. The initial graph in $\boldsymbol{X}$ is defined as $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \boldsymbol{A}^{(t)})$, where $\mathcal{V} = \{\boldsymbol{X}_{1,:}, \boldsymbol{X}_{2,:}, \ldots, \boldsymbol{X}_{w^2,:}\}$ and $\mathcal{E}$ are the initial node and edge sets. $\boldsymbol{A}^{(t)} \in \mathbb{R}^{w^2\times w^2}$ is the initial fluctuating adjacent

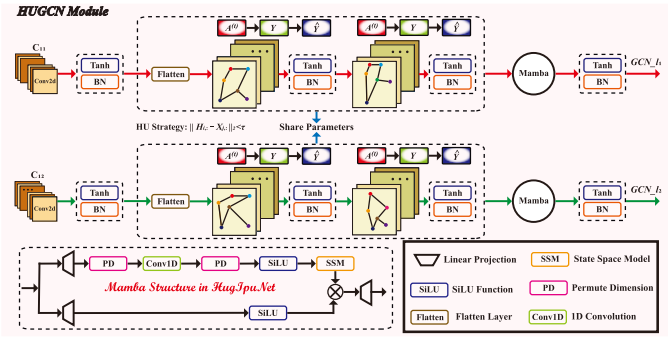


Fig. 3. Architecture of the HUGCN module. The shared layers are marked by blue lines in these two unique streams.

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matrix for $\mathcal{G}$ and is formulated as

$$\boldsymbol{A}_{i,j}^{(t)} = \exp\left(-\frac{\left\|\boldsymbol{X}_{i,:} - \boldsymbol{X}_{j,:}\right\|_2^2}{2\left(\sigma + \sin\left(\beta t\right)\right)^2}\right) \tag{2}$$

where $\sigma = 1$ and $\beta = 0.1$, t is the training time and is denoted as the training epoch number. In particular, $\sigma + \sin(\beta t)$ aims to introduce uncertainty when constructing the dynamic adjacent matrix. This approach limits the effect caused by the distortion of on-ground object descriptions due to complex environmental scenarios. Specifically, this highly robust version effectively calculates the internode correlation in complex environmental scenarios, obtaining relatively high matrix elements when the spectral curves of two samples exhibit similarity. Subsequently, both $\boldsymbol{X}$ and $\boldsymbol{A}^{(t)}$ are delivered into the first HUGCN stage for generating the intermediate output $\boldsymbol{Y}$.

Afterward, the intermediate graph in $\boldsymbol{Y}$ can be denoted as $\mathcal{G}' = (\mathcal{V}', \mathcal{E}', \boldsymbol{A}'^{(t)})$, where $\mathcal{V}' = \{\boldsymbol{Y}_{1,:}, \boldsymbol{Y}_{2,:}, \ldots, \boldsymbol{Y}_{w^2,:}\}$ and $\mathcal{E}'$ are the intermediate node and edge sets. $\boldsymbol{A}'^{(t)}$ is the intermediate fluctuating adjacent matrix. To avoid losing information inevitably during the graph construction procedure, one HU strategy is presented in the second HUGCN stage for updating $\boldsymbol{Y}$, which is formulated as

$$\hat{\boldsymbol{Y}}_{i,:} = \begin{cases} \boldsymbol{X}_{j,:}, & \text{if } \boldsymbol{\Psi}_{i,j} \le \tau \\ \boldsymbol{Y}_{i,:}, & \text{otherwise} \end{cases} \tag{3}$$

where $\boldsymbol{\Psi}_{i,j} = \| \boldsymbol{Y}_{i,:} - \boldsymbol{X}_{j,:} \|_2^2$ measures the distance between $\boldsymbol{Y}_{i,:}$ and $\boldsymbol{X}_{j,:}$, $\hat{\boldsymbol{Y}}$ is the updated output, τ is one threshold, $1 \le j \le w^2$, and all $\| \boldsymbol{Y}_{i,:} - \boldsymbol{X}_{j,:} \|_2^2$ values are normalized into $[0, 1]$. This HU strategy replaces some intermediate nodes with corresponding initial nodes when their differences are minimal (less than τ), thereby preserving the graph construction accuracy and avoiding inevitable information loss. To improve smoothness, (3) can be rewritten as

$$\hat{\boldsymbol{Y}}_{i,:} = \begin{cases} \left(1 - \boldsymbol{\Psi}_{i,j}\right) \boldsymbol{X}_{j,:} + \boldsymbol{\Psi}_{i,j} \boldsymbol{Y}_{i,:}, & \text{if } \boldsymbol{\Psi}_{i,j} \le \tau \\ \boldsymbol{Y}_{i,:}, & \text{otherwise.} \end{cases} \tag{4}$$

Compared to (3), (4) softly balances $\boldsymbol{X}$ and $\boldsymbol{Y}$ by constructing $\boldsymbol{\Psi}$ to enhance the feature smoothness.

The calculation procedure when generating $\boldsymbol{Y}$ by $\boldsymbol{X}$ and $\boldsymbol{A}^{(t)}$ is provided. The intermediate output $\boldsymbol{Y}$ is generated as

$$\mathbf{Y} = \text{Tanh}\left(\mathbf{D}^{-\frac{1}{2}}\left(\mathbf{A}^{(t)} + \mathbf{I}\right)\mathbf{D}^{-\frac{1}{2}}\mathbf{X}\mathbf{W}\right) \tag{5}$$

where $\mathbf{D}$ is the degree matrix of $\mathbf{A}^{(t)} + \mathbf{I}$, $\mathbf{I}$ is the identity matrix, and $\mathbf{W}$ is one trainable matrix. The HUGCN layer number is fixed in the HUGCN module since using excessive layers tends to result in an overly large aggregated view field, making it difficult to extract useful information. The outputs of the HUGCN module are GCN_$\mathbf{l}_1$ and GCN_$\mathbf{l}_2$, respectively.

C. SDL Module

The SDL module is presented in Fig. 4, which aims to extract high-quality HS features from dual perspectives and limit the effect of complex environmental illumination on spectrum distributions. This module consists of two streams: similar to the HUGCN module, the first stream considers two stacked HUGCN layers to robustly limit the effect caused by

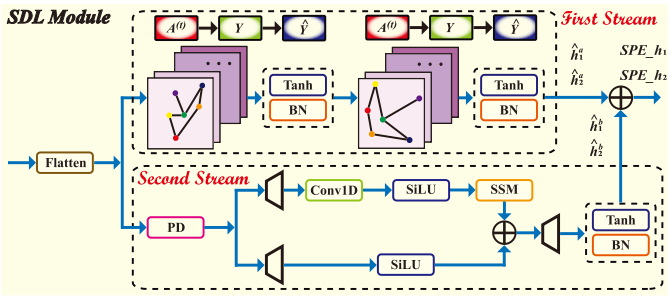


Fig. 4. Architecture of the SDL module with two streams to obtain spectral information comprehensively.

on-ground object spectrum distortion due to complex environmental illumination; inspired by Mamba structures, the second stream consists of one SSM-based structure for excellent performance on long-range dependency enhancement and global information analysis. The two inputs $\overline{\mathbf{h}}_1$ and $\overline{\mathbf{h}}_2$ are loaded into both streams in the SDL module. The structures for $\overline{\mathbf{h}}_1$ and $\overline{\mathbf{h}}_2$ share the same SDL parameters. These inputs are reshaped into $[\text{BS}, w^2, c]$ since their shapes are both $[\text{BS}, w, w, c]$ initially.

1) The HUGCN layers in the first stream are similar to those in the HUGCN module. The outputs of this stream are denoted as $\hat{\mathbf{h}}_1^a$ and $\hat{\mathbf{h}}_2^a$, respectively.
2) The detailed structure of the second stream is shown in Fig. 2. For a clear description, its input is denoted as $\hat{\mathbf{h}}$ directly. Initially, one permute dimension (PD) layer changes the shape of $\hat{\mathbf{h}}$ into $[\text{BS}, c, w^2]$. Notably, the linear projection (LP) layers, the 1-D convolution (Conv1d) layers with w^2 nodes, and the SiLU functions do not change the latent feature shape. The SSM calculation procedure [17] can be formulated as

$$\begin{aligned} \hat{\mathbf{p}}^{(t)} &= \overline{\mathbf{P}}\hat{\mathbf{p}}^{(t-1)} + \overline{\mathbf{Q}}\hat{\mathbf{h}}^{(t)} \\ \hat{\mathbf{p}}^{(t)b} &= \mathbf{T}\hat{\mathbf{p}}^{(t)} \\ \overline{\mathbf{P}} &= \exp\left(\Delta\mathbf{P}\right) \\ \overline{\mathbf{Q}} &= \left(\Delta\mathbf{P}\right)^{-1}\left(\exp\left(\Delta\mathbf{P}\right) - \mathbf{I}\right)\Delta\mathbf{Q} \end{aligned} \tag{6}$$

where $\hat{\mathbf{p}}^{(t)}$ and $\hat{\mathbf{p}}^{(t)b}$ are the states of $\hat{\mathbf{p}}$ and $\hat{\mathbf{p}}^b$ in the tth epoch, $\hat{\mathbf{p}}^b$ is the output, $\mathbf{P}$, $\mathbf{Q}$ and $\mathbf{T}$ are trainable matrices, Δ is a timescale parameter, $\overline{\mathbf{P}}$, and $\overline{\mathbf{Q}}$ are obtained by a fixed zero-order hold (ZOH) technique. Afterward, the whole optimization procedure can be rewritten as

$$\begin{aligned} \overline{\mathbf{K}} &= \left(\mathbf{T}\overline{\mathbf{Q}}, \mathbf{T}\overline{\mathbf{P}\mathbf{Q}}, \ldots, \mathbf{T}\overline{\mathbf{P}^{c-1}\mathbf{Q}}\right) \\ \hat{\mathbf{p}}^b &= \hat{\mathbf{h}} * \overline{\mathbf{K}} \end{aligned} \tag{7}$$

where $\overline{\mathbf{K}}$ is a convolutional kernel. The outputs of this stream are denoted as $\hat{\mathbf{h}}_1^b$ and $\hat{\mathbf{h}}_2^b$, respectively. Finally, the outputs of the SDL module are SPE_$\mathbf{h}_1 = \hat{\mathbf{h}}_1^a + \hat{\mathbf{h}}_1^b$ and SPE_$\mathbf{h}_2 = \hat{\mathbf{h}}_2^a + \hat{\mathbf{h}}_2^b$, respectively, for the two inputs. The SDL module attempts to unify global illumination-wise property attributes by analyzing a large amount of spatial–spectral information. The SSM-related structure captures interchannel relationships accurately by dynamically selecting and forgetting information from the previous stage.

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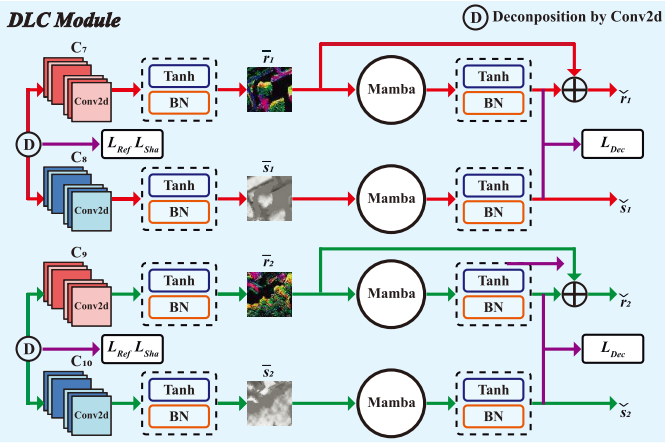


Fig. 5. Architecture of the DLC module. Two patch pairs are randomly selected for training since this module needs to calculate the reflectance and shading gradients.

D. DLC Module

The DLC module in Fig. 5 aims to decompose the HSIs into one reflectance component and one shading component with the help of DSMs to increase fusion interpretability [29], [37], [38]. HugIpuNet fully harnesses the potential of both the reflectance and shading components, avoiding the information loss caused by neglecting the latter directly. Environmental illumination can be influenced by multiple factors in real-world tasks, for example, solar elevation angle and ambient occlusion. In this work, these factors are not considered for simplification because the target region areas are limited, meaning that these factors are generally consistent. The input pairs are $[\bar{\mathbf{h}}_3, \bar{\mathbf{l}}_1]$ and $[\bar{\mathbf{h}}_4, \bar{\mathbf{l}}_2]$. The structures for both input pairs are the same. Inspired by (1), $\bar{\mathbf{h}}_3$ can be decomposed into the reflectance component $\bar{\mathbf{r}}_1$ and the shading component $\bar{\mathbf{s}}_1$ by two Conv2d layers, respectively. The decomposition procedure should minimize the constraint as follows:

$$L_{\text{Dec}} = \frac{1}{c}\sum_{i=1}^{c}\left[\text{GAP}\left(\bar{\mathbf{h}}_3 - \bar{\mathbf{r}}_1 \circ \bar{\mathbf{s}}_1\right]_i^2 \tag{8}$$

where GAP$(\cdot)$ is the global average pooling layer. Similarly, $\bar{\mathbf{h}}_4$ is decomposed into $\bar{\mathbf{r}}_2$ and $\bar{\mathbf{s}}_2$. Moreover, this module introduces two specific constraints on both $\mathbf{R}$ and $\mathbf{S}$.

1) Constraint on $\mathbf{R}$*:* HugIpuNet assumes that the differences among on-ground objects can be reflected by variations in their spectral curves as well as by their spatial arrangement distances. Therefore, the reflectance gradient between $\bar{\mathbf{h}}_3$ and $\bar{\mathbf{h}}_4$ can be formulated as

$$\begin{aligned}\mathbf{g}^R &= \left\{\mathbf{g}_1^R, \mathbf{g}_2^R, \ldots, \mathbf{g}_c^R\right\}\\ \mathbf{g}_i^R &= \left|\text{GAP}\left(\bar{\mathbf{h}}_3\right) - \text{GAP}\left(\bar{\mathbf{h}}_4\right)\right|_i^{-1} \cdot \text{Euc}\left(\bar{\mathbf{h}}_3, \bar{\mathbf{h}}_4\right)^{-1}\end{aligned} \tag{9}$$

where Euc$(\cdot)$ and $|\cdot|$ represent the Euclidean distance and the absolute value operator, $1 \le i \le c$. Thus, the decomposition procedure should minimize the following constraint:

$$L_{\text{Ref}} = \frac{1}{c}\sum_{i=1}^{c}\left\{\text{Sig}\left(\mathbf{g}_i^R\right)\left[\text{GAP}\left(\bar{\mathbf{r}}_1\right) - \text{GAP}\left(\bar{\mathbf{r}}_2\right)\right]_i^2\right\} \tag{10}$$

where Sig$(\cdot)$ is the Sigmoid function for normalization.

2) Constraint on $\mathbf{S}$*:* HugIpuNet further assumes that objects with similar elevations always have similar environmental illumination, which connects HSIs and LiDAR-derived DSMs from the geological perspective. This assumption is feasible when analyzing regions with limited areas and neglecting tiny shadows along contour edges. Specifically, the shading gradient between $\bar{\mathbf{l}}_1$ and $\bar{\mathbf{l}}_2$ can be formulated as

$$\begin{aligned}\mathbf{g}^S &= \left\{\mathbf{g}_1^S, \mathbf{g}_2^S, \ldots, \mathbf{g}_c^S\right\}\\ \mathbf{g}_i^S &= \left|\text{GAP}\left(\bar{\mathbf{l}}_1\right) - \text{GAP}\left(\bar{\mathbf{l}}_2\right)\right|_i^{-1} \cdot \text{Euc}\left(\bar{\mathbf{l}}_1, \bar{\mathbf{l}}_2\right)^{-1}.\end{aligned} \tag{11}$$

The elevation differences in the adjacent noncontour area are always minimal, ensuring a continuous shading gradient and facilitating the continuous $\mathbf{S}$ calculation. Afterward, the decomposition procedure should minimize the constraint as

$$L_{\text{Sha}} = \frac{1}{c}\sum_{i=1}^{c}\left\{\text{Sig}\left(\mathbf{g}_i^S\right)\left[\text{GAP}\left(\bar{\mathbf{s}}_1\right) - \text{GAP}\left(\bar{\mathbf{s}}_2\right)\right]_i^2\right\}. \tag{12}$$

After generating high-quality components by decomposition, the reflectance components (i.e., $\bar{\mathbf{r}}_1$ and $\bar{\mathbf{r}}_2$) and the shading components (i.e., $\bar{\mathbf{s}}_1$ and $\bar{\mathbf{s}}_2$) are separately delivered into Mamba-based residual stages for information stabilization. The reason why using the shading components is that they contain important object arrangement information in the spatial domain and provide supplementary geometric information for LiDAR-derived DSM. The structures of these stages are provided in Fig. 2. Furthermore, the processed reflectance components (i.e., $\check{\mathbf{r}}_1$ and $\check{\mathbf{r}}_2$) and the processed shading components (i.e., $\check{\mathbf{s}}_1$ and $\check{\mathbf{s}}_2$) are regarded as the outputs of the DLC Module.

E. Classification and Optimization

The generated features GCN_$\mathbf{l}_1$, SPE_$\mathbf{h}_1$, $\check{\mathbf{r}}_1$, and $\check{\mathbf{s}}_1$ for the first patch pair are concatenated for classification. The generated features GCN_$\mathbf{l}_2$, SPE_$\mathbf{h}_2$, $\check{\mathbf{r}}_2$, and $\check{\mathbf{s}}_2$ for the second patch pair are preserved to calculate the constraints. Considering the two patch pairs are randomly selected, this setting is feasible. The concatenated feature sequentially passes through one global average pooling layer, one fully connected (FC) layer, and one Softmax activation layer, ultimately obtaining the predicted labels. The loss function can be formulated as

$$L = \alpha\left(L_{\text{Dec}} + L_{\text{Ref}} + L_{\text{Sha}}\right) + L_C \tag{13}$$

where L_C measures the cross-entropy between the predicted labels and the real labels, and α is a loss weight to control the trend of intrinsic decomposition. The computing complexity of HugIpuNet is $\mathcal{O}(w^2c^2)$, where w is the sliding window side and c is the HS channel number. The detailed training time can be affected by several factors, for example, the epoch number t and GPU capabilities.

IV. Experimental Results and Analysis

A. Dataset Description

The Muufl Gulfport (MUUFL) dataset was captured over the University of Southern Mississippi Gulf Park Campus, Mississippi, USA Its HSI was recorded by the compact airborne spectrographic imager (CASI) 1500 sensor and has 64 spectral

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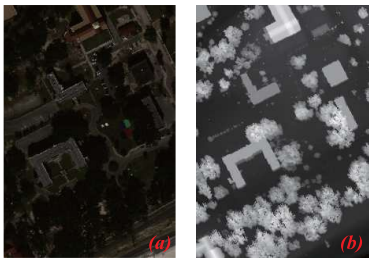


Fig. 6. Visualization of the MUUFL dataset, including (a) HS pseudo color image and (b) LiDAR-derived DSM gray image.

TABLE I
TRAINING AND TESTING SAMPLE NUMBERS IN THE MUUFL DATASET

Class	Land-cover Category	Training	Testing	Total
1	Trees	232	23014	23246
2	Mostly Grass	42	4228	4270
3	Mixed Ground Surface	69	6813	6882
4	Dirt and Sand	18	1808	1826
5	Road	67	6620	6687
6	Water	5	461	466
7	Building Shadow	22	2211	2233
8	Building	62	6178	6240
9	Sidewalk	14	1371	1385
10	Yellow Curb	2	181	183
11	Cloth Panels	3	266	269

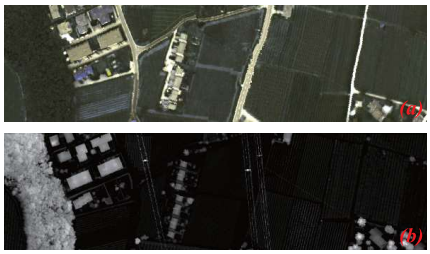


Fig. 7. Visualization of the Trento dataset, including (a) HS pseudo color image and (b) LiDAR-derived DSM gray image.

TABLE II
TRAINING AND TESTING SAMPLE NUMBERS IN THE TRENTO DATASET

Class	Land-cover Category	Training	Testing	Total
1	Apple Trees	40	3994	4034
2	Buildings	29	2874	2903
3	Ground	5	474	479
4	Wood	91	9032	9123
5	Vineyard	105	10396	10501
6	Roads	32	3142	3174

channels ranging from 375 to 1050 nm. The shapes and spatial resolutions of both its HSI and DSM are 325 × 220 and 1 m, respectively. The sample numbers in all 11 categories are provided in Table I. The HSI and DSM visualization is shown in Fig. 6.

The Trento dataset was captured over the south of Trento, Italy. Its HSI was recorded by the AISA Eagle sensor and contains 63 spectral channels ranging from 402.89 nm to

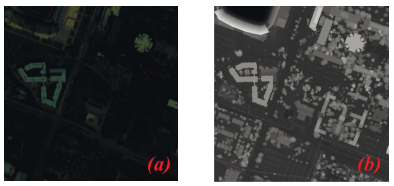


Fig. 8. Visualization of the Houston dataset, including (a) HS pseudo color image and (b) LiDAR-derived DSM gray image.

TABLE III
TRAINING AND TESTING SAMPLE NUMBERS IN THE HOUSTON DATASET

Class	Land-cover Category	Training	Testing	Total
1	Healthy Grass	14	1439	1453
2	Stressed Grass	51	5020	5071
3	Evergreen Trees	64	6371	6435
4	Deciduous Trees	30	2925	2955
5	Residential Buildings	81	7982	8063
6	Non-Residential Buildings	482	47782	48264
7	Roads	70	6969	7039
8	Sidewalks	125	12372	12497
9	Major Thoroughfares	179	17722	17901
10	Paved Parking Lots	42	4114	4156
11	Cars	11	1090	1101
12	Stadium Seats	68	6756	6824

989.09 nm. Its DSM was recorded by the Optech airborne laser terrain mapper (ALTM) 3100EA sensor. Their shapes and spatial resolutions are 166 × 600 and 1 m, respectively. The sample numbers in all six categories are provided in Table II. The HSI and DSM visualization is shown in Fig. 7.

The Houston dataset was captured over the Houston University campus by the National Center for Airborne Laser Mapping (NCALM) sensor. Its HSI has 48 spectral channels ranging from 380 to 1050 nm. The spatial resolutions of its HSI and DSM are 1 m and 50 cm, respectively. An interval sampling strategy was utilized to unify their shapes. Notably, one partitioned subscene with 601 × 596 samples (i.e., UH_NAD83_272652_3289689) was chosen since the whole dataset scale is large. The sample numbers in all 12 categories are provided in Table III. The HSI and DSM visualization is shown in Fig. 8.

To evaluate category sizes, the sample ratio is defined to calculate the ratio of the sample number in one category to the total sample number. The categories with less than 2‰ sample ratios were removed. Moreover, the training set is generated by randomly choosing 1% samples in each category. The remaining 99% samples generate the testing set.

B. Experimental Setting

The whole experiments were conducted on one PC with the Intel[1] Core[2] i7-10750H CPU and with the NVIDIA GeForce GTX 1650 GPU. The codes were carried out in the Keras framework. The adaptive moment estimation (Adam) was selected for network optimization. The loss weight α

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was set as 0.0001. The learning rate was set as 5×10^{-3}. The batch size was chosen within $\{64, 128, 256, 512\}$. The sliding window side w was chosen within $\{3, 5, 7, 9, 11\}$. The experiments were conducted ten times to mitigate the influence of random selection bias. The optimization procedures would stop if all network parameters become stable. Five evaluation metrics were selected, including the accuracy per category, overall accuracy (OA), average accuracy (AA), precision (PR), and $F1$-score ($F1$). Fourteen state-of-the-art methods were selected for comparison, including three baselines (i.e., early fusion, middle fusion, and late fusion) [39], ENDnet [39], two CNN-based networks (i.e., CNN_H and CNN_L) [40], multiview hierarchical network (MVHN) [41], multiscale self-calibrated convolution (MSNetSC) [42], local information interaction transformer (LIIT) [43], morphological convolution and attention calibration network (MCAC) [44], spatial–spectral saliency reinforcement network (Sal^2RN) [45], hashing-based deep metric learning (HDML) [46], MambaHSI [17], and one graph structure and vision Mamba-based network (GraphMamba) [36]. Notably, considering that MambaHSI and GraphMamba are single-modal methods, their structures were adjusted by adding streams for LiDAR-derived DSM, which are consistent with their HS streams. The structures and parameters of all comparison methods were fine-tuned for a fair comparison. The method descriptions are listed as follows.

1) ***ENDnet*** uses FC layers to reconstruct its inputs without using spatial information.
2) ***CNN_H*** and ***CNN_L*** share similar structures and contain stacked CNN layers and one Softmax-activated FC layer.
3) ***MVHN*** integrates principal components analysis with Gabor filters for cross-modal feature extraction and introduces a stacking mechanism-based residual module for fusing HS and DSM features.
4) ***MSNetSC*** considers a self-calibrated convolution motivated hierarchical residual structure for satisfactory performance. The network contains multiple self-attention mechanisms.
5) ***LIIT*** is one multihead self-attention-based transformer for local information interaction.
6) ***MCAC*** is a morphological convolution and attention calibration-based network to classify HS and DSM samples.
7) ***Sal^2RN*** integrates one cross-layer interaction module with one customized center spectrum correction module for spatial–spectral information.
8) ***HDML*** learns high-discrimination fused features through attention motivated feature fusion modules for capturing both single-modal and cross-modal features.
9) ***MambaHSI*** contains spectral and spatial Mamba blocks for HSIs and DSMs, respectively, to achieve satisfactory fusion performance.
10) ***GraphMamba*** balances GCN layers with Mamba-based structures for feature extraction. The network structures for HSIs and DSMs correspond to the detailed RS modality shapes.

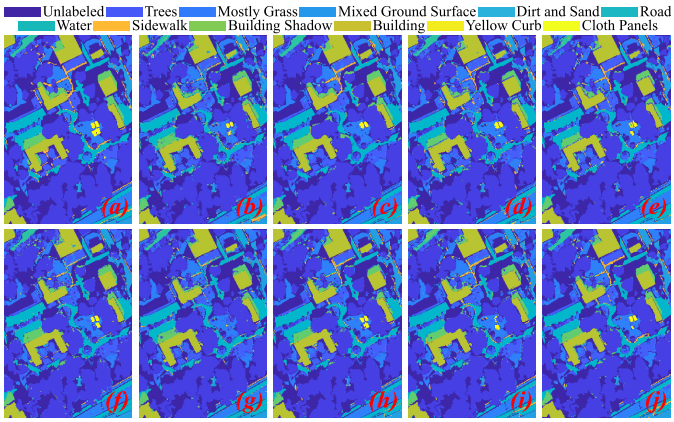


Fig. 9. (a) Ground truth and (b)–(j) classification maps on the MUUFL dataset, including (b) MVHN, (c) MSNetSC, (d) LIIT, (e) MCAC, (f) Sal^2RN, (g) HDML, (h) MambaHSI, (i) GraphMamba, and (j) HugIpuNet.

C. Experimental Results and Comparison

Experiments were conducted on these three datasets to prove the effectiveness of HugIpuNet. The classification accuracies for these datasets are presented in Tables IV–VI, respectively. The best accuracies are highlighted in bold. Generally, pixel-wise techniques exhibit inferior performance because of their inability to capture spatial information comprehensively. Considering that the experimental results on the first two datasets clearly verify the shortcomings of these pixel-wise techniques, these techniques are not selected for comparison on the Houston dataset. Furthermore, cross-modal methods generally generate superior classification performance over single-modal methods, that is, CNN_H and CNN_L, further verifying that fusing both HSIs and DSMs is feasible. Among all comparison methods, HugIpuNet achieves the highest OAs across all three datasets, surpassing the second best techniques by 1.37% (MUUFL dataset), 0.42% (Trento dataset), and 1.67% (Houston dataset). On the MUUFL dataset, HugIpuNet also obtains significantly improved PR compared to other approaches. This improvement comes from its multimechanism-based structure, which emphasizes collaborative feature interactions rather than isolated feature extractions. The capability of HugIpuNet in complicated environmental scenarios is clearly validated on these three datasets. Notably, the tenth category (*Yellow Curb*) and the 11st category (*Cloth Panels*) in the MUUFL dataset contain limited training samples. Thus, the samples in these two land-cover categories cannot be classified accurately by most techniques. The reason is that limited training samples cannot provide enough information for network optimization and impede effective global–local feature extraction by GCN layers.

The classification maps on these three datasets are presented in Figs. 9–11, respectively. The maps prove that HugIpuNet achieves superior accuracy across most land-cover categories. For example, the *mixed ground surface* and *water* samples in the MUUFL dataset can be clearly identified by HugIpuNet. The *ground* samples in the Trento dataset can also be accurately classified by HugIpuNet, although there are only five *ground* training samples. In the Houston dataset,

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TABLE IV

CLASSIFICATION ACCURACIES, INCLUDING OA (%), AA (%), PR (%), AND $F1$ ($\times 100$), ON THE MUUFL DATASET

Class	Early	Middle	Late	ENDnet	CNN_H	CNN_L	MVHN	MSNetSC	LIIT	MCAC	Sal^{2}RN	HDML	MambaHSI	GraphMamba	Our
1	94.42	94.36	92.64	93.75	96.57	84.27	96.58	96.20	96.72	**97.74**	95.28	97.21	97.56	97.29	97.21
2	51.40	64.10	51.49	71.14	63.55	33.96	66.75	74.01	66.91	69.70	73.89	69.37	71.26	**77.37**	74.79
3	82.03	65.60	74.08	66.74	76.12	35.40	73.36	81.08	79.14	83.21	80.41	77.78	81.37	81.39	**83.55**
4	69.03	73.34	80.92	47.51	59.02	4.48	56.80	**91.54**	75.88	79.76	88.44	77.71	77.16	84.57	85.29
5	93.23	86.72	91.10	88.94	93.04	62.81	83.08	92.66	91.47	93.69	94.64	90.95	**95.47**	93.70	94.59
6	72.89	83.73	73.97	82.00	45.55	74.40	41.00	95.66	89.15	59.22	85.03	63.12	48.81	51.41	**95.88**
7	47.04	64.18	62.96	67.62	62.60	15.65	65.08	**81.28**	69.74	70.87	77.02	75.35	64.63	69.52	77.97
8	86.73	90.84	88.94	91.19	95.03	76.17	93.40	97.17	95.60	95.00	95.40	96.62	**97.65**	96.65	97.38
9	57.91	50.40	**61.05**	47.78	41.58	2.41	17.29	34.21	32.53	45.15	33.77	45.73	41.36	43.03	57.18
10	24.86	**28.73**	25.41	15.47	0	0.55	2.21	0	25.41	7.18	0.55	3.87	14.92	0.56	0.66
11	41.73	59.40	56.39	**87.97**	66.92	0	42.11	40.98	45.11	45.11	59.02	31.58	62.03	20.68	43.98
OA	83.90	83.31	83.35	83.35	85.67	61.92	83.40	89.18	87.26	88.83	88.43	87.82	88.75	89.05	**90.55**
AA	65.57	69.22	69.00	69.10	63.63	35.46	57.97	71.34	69.79	67.88	71.22	66.30	68.38	65.11	**74.04**
PR	75.19	68.96	73.87	68.61	72.55	37.56	65.35	71.65	72.95	77.45	77.88	81.11	79.43	80.53	**87.61**
F1	68.40	68.43	70.55	68.71	67.02	35.21	60.62	72.50	70.75	71.62	73.23	69.83	72.51	68.00	**76.68**

TABLE V

CLASSIFICATION ACCURACIES, INCLUDING OA (%), AA (%), PR (%), AND $F1$ ($\times 100$), ON THE TRENTO DATASET

Class	Early	Middle	Late	ENDnet	CNN_H	CNN_L	MVHN	MSNetSC	LIIT	MCAC	Sal^{2}RN	HDML	MambaHSI	GraphMamba	Our
1	66.85	64.37	82.37	69.23	99.10	89.26	97.45	92.21	96.19	93.16	98.60	98.97	**99.57**	**99.57**	98.40
2	97.63	95.09	97.22	95.06	89.98	79.75	92.14	97.77	94.33	**99.10**	94.89	91.75	97.43	98.40	97.56
3	91.14	93.88	86.29	78.69	94.73	13.71	87.76	90.08	98.10	91.98	86.92	92.83	98.59	80.17	**99.37**
4	99.25	98.77	99.07	98.48	99.35	92.36	99.68	98.49	99.79	**100.00**	99.97	99.86	99.97	99.88	**100.00**
5	90.04	91.84	88.27	87.17	99.84	98.02	98.03	96.50	99.90	**100.00**	98.86	98.94	97.04	97.67	99.44
6	89.88	90.80	87.62	91.15	77.59	76.89	87.11	92.17	97.58	92.78	93.41	96.94	97.36	**98.12**	96.59
OA	90.46	90.50	91.51	89.23	96.23	89.83	96.57	96.10	98.57	98.11	98.01	98.22	98.36	98.43	**98.99**
AA	89.13	89.13	90.14	86.63	93.43	75.00	93.69	94.54	97.65	96.17	95.44	96.55	98.38	95.64	**98.56**
PR	88.77	90.52	88.71	88.16	91.74	81.87	93.84	92.81	95.68	98.01	97.23	97.25	97.97	98.09	**98.20**
F1	88.85	89.72	89.26	87.36	92.31	76.40	93.74	93.60	96.52	97.03	96.25	96.86	98.15	96.66	**98.38**

TABLE VI

CLASSIFICATION ACCURACIES, INCLUDING OA (%), AA (%), PR (%), AND $F1$ ($\times 100$), ON THE HOUSTON DATASET

Class	MVHN	MSNetSC	LIIT	MCAC	Sal^{2}RN	HDML	MambaHSI	GraphMamba	Our
1	75.82	78.39	50.73	60.81	71.16	**83.60**	80.82	78.04	73.59
2	83.17	77.03	74.96	78.41	**89.44**	83.82	81.99	84.62	86.67
3	75.53	85.73	86.38	89.70	95.21	96.88	97.39	97.49	**97.99**
4	78.36	82.43	84.68	62.19	83.86	**87.69**	86.80	79.25	85.13
5	79.27	76.35	**95.16**	85.40	86.22	91.73	86.34	86.76	91.29
6	96.17	93.27	93.89	95.02	96.81	97.01	96.85	96.03	**98.00**
7	38.79	36.65	44.08	36.05	46.85	44.64	45.99	42.85	**54.64**
8	65.71	68.63	58.97	60.04	56.98	71.86	70.15	69.55	**75.06**
9	76.83	72.23	78.15	82.50	**89.74**	86.60	84.84	87.25	85.46
10	87.68	86.63	89.57	83.13	89.67	91.61	**93.32**	91.98	92.32
11	57.52	29.27	54.40	59.54	51.38	28.26	63.39	63.30	**79.17**
12	93.18	94.46	96.05	97.57	98.08	96.48	**99.22**	96.46	97.99
OA	82.65	81.00	82.89	82.78	86.50	87.74	87.39	86.90	**89.41**
AA	75.67	73.42	75.59	74.20	79.62	80.02	82.26	81.13	**84.78**
PR	75.89	75.03	78.54	77.28	81.22	85.24	84.80	84.07	**86.95**
F1	75.42	73.15	76.46	75.43	80.03	81.16	83.26	82.20	**85.75**

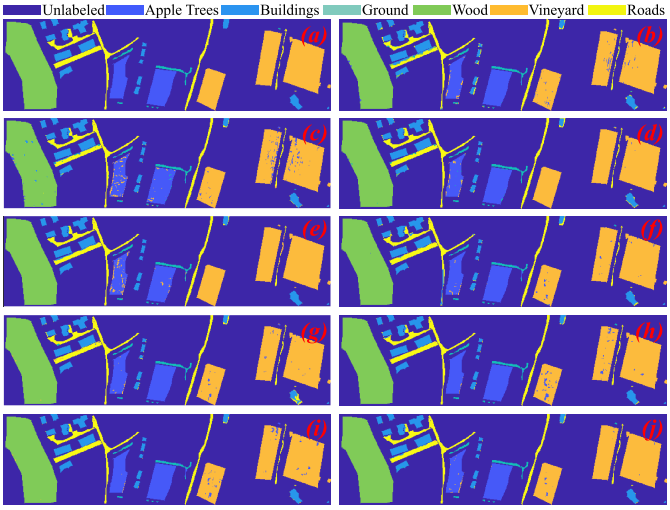


Fig. 10. (a) Ground truth and (b)–(j) classification maps on the Trento dataset, including (b) MVHN, (c) MSNetSC, (d) LIIT, (e) MCAC, (f) Sal^{2}RN, (g) HDML, (h) MambaHSI, (i) GraphMamba, and (j) HugIpuNet.

the *nonresidential buildings* samples can be precisely detected, demonstrating that HugIpuNet is capable of mitigating environmental interference on building contours. Moreover, the classification maps of HugIpuNet contain less scattered noise than other comparison techniques.

Fig. 12 presents the visualization of the reflectance and shading components decomposed by HugIpuNet, further verifying that HugIpuNet is feasible to enhance the joint classification interpretability. It shows that the reflectance component effectively captures the on-ground object materials in uniform illumination. Meanwhile, the shading component specifically highlights the illumination variations rather than the detailed materials. The results on these three datasets clearly show that the shading components are correlated with the LiDAR elevations, which is consistent with human cognition. Furthermore, the shading components also record

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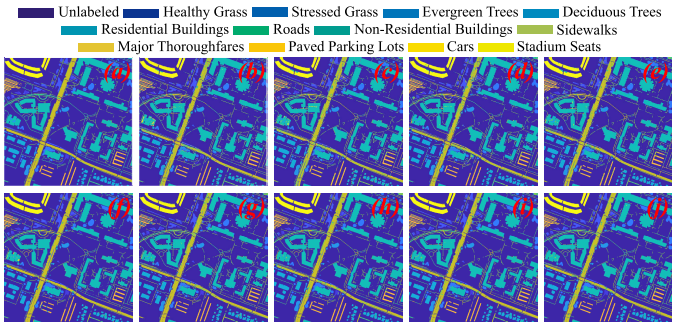


Fig. 11. (a) Ground truth and (b)–(j) classification maps on the Houston dataset, including (b) MVHN, (c) MSNetSC, (d) LIIT, (e) MCAC, (f) Sal^{2}RN, (g) HDML, (h) MambaHSI, (i) GraphMamba, and (j) HugIpuNet.

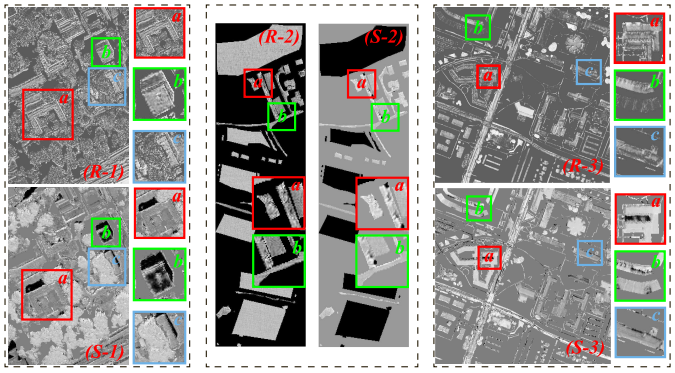


Fig. 12. Decomposition performance of HugIpuNet on three datasets, (R-*) and (S-*) correspond to the reflectance and shading components, (*-1)-(*-3) correspond to the MUUFL, Trento, and Houston datasets, some typical regions are squared and magnified.

important discriminant features, which contribute to enhancing the joint classification performance. Thus, HugIpuNet comprehensively utilizes both components rather than neglecting either one.

D. Ablation Study and Parameter Selection

For a clear description and comparison, both the ablation and parameter selection experiments were conducted on the MUUFL dataset in this article.

1) Necessity of Both RS Modalities: To verify the necessity of considering both HSIs and LiDAR-derived DSMs in joint classification tasks, several HugIpuNet-based alternative scenarios were provided by using different modality pairs as follows:

1) **HS + ∅**: only one HSI is chosen as the input, the LiDAR-related structures are removed.
2) **LiDAR + ∅**: only one LiDAR-derived DSM is chosen as the input, and the HS-related structures are removed.
3) **HS + HS**: the inputs are both HSIs.
4) **LiDAR + LiDAR**: the inputs are both DSMs.
5) **HS + LiDAR** (the presented structure): the inputs are one HSI and one LiDAR-derived DSM, respectively.

Their classification accuracies are provided in Table VII, showing that fusing both RS modalities for joint classification is feasible and capable of increasing the land-cover identification ability. The ∅ stands for the null set, which is identified as only using HSIs or DSMs. To be specific, the OAs of HugIpuNet on these three datasets are 3.47%, 3.11%, and 3.97% larger than those of HS + ∅ and are 29.28%, 13.12%, and 26.05% larger than those of LiDAR + ∅. In other words, when only using HSIs, all accuracies are lower than those of HS + LiDAR (the presented structure) on the three datasets. When only using DSMs, the classification performance is not satisfactory. Therefore, HSIs can provide more discriminant information than DSMs, and DSMs record essential elevations as supplementary information for interpretation. Furthermore, the results also show that fusing two identical RS modalities cannot improve the classification performance.

TABLE VII

CLASSIFICATION ACCURACIES, INCLUDING OA (%), AA (%), PR (%), AND $F1$ ($\times 100$), ON THE THREE DATASETS TO PROVE THE NECESSITY OF BOTH RS MODALITIES

Datasets	Scenarios	Metrics			
		OA (%)	AA (%)	PR (%)	F1 ($\times$100)
MUUFL	HS+∅	87.08	70.69	83.39	73.36
	LiDAR+∅	61.27	25.93	32.92	24.63
	HS+HS	87.54	67.10	84.39	71.05
	LiDAR+LiDAR	60.65	25.27	28.04	24.12
	HS+LiDAR	**90.55**	**74.01**	**87.61**	**76.68**
Trento	HS+∅	95.88	94.22	95.01	94.61
	LiDAR+∅	85.87	74.54	75.40	74.90
	HS+HS	94.30	93.42	94.21	93.77
	LiDAR+LiDAR	87.60	71.14	81.10	71.91
	HS+LiDAR	**98.99**	**98.56**	**98.20**	**98.38**
Houston	HS+∅	85.44	76.38	82.67	76.94
	LiDAR+∅	63.36	32.72	37.95	31.23
	HS+HS	85.72	76.61	81.67	77.82
	LiDAR+LiDAR	63.14	31.83	32.82	29.48
	HS+LiDAR	**89.41**	**84.78**	**86.95**	**85.75**

2) Necessity of Network Modules: To prove the necessity of using all network modules in HugIpuNet, diverse alternative structures were constructed by removing partial modules from HugIpuNet, which are listed as:

CPS-related structures.

1) K_1: removes the whole CPS module from HugIpuNet.

HUGCN-related structures.

2) K_2: removes the whole HUGCN module from the HugIpuNet structure and preserves the other structures.
3) K_3: replaces all HUGCN layers by classic GCN layers.
4) K_4: fixes $(\sigma + \sin(\beta t))^2$ to 1 in (2).

SDL-related structures.

5) K_5: removes the whole SDL module from HugIpuNet.
6) K_6: removes all HUGCN layers from the first stream in the SDL module.
7) K_7: removes all SSM and Mamba-related layers from the second stream in the SDL module.

DLC-related structures.

8) K_8: removes the whole DLC module from HugIpuNet.
9) K_9: preserves the reflectance component and removes the shading component in HugIpuNet.
10) K_{10}: preserves the shading component and removes the reflectance component in HugIpuNet.
11) K_{11}: removes all SSM and Mamba-related layers from the DLC module.
12) K_{12}: replaces the concatenation layer with the addition layer for information fusion in HugIpuNet.

Their classification accuracies are all presented in Table VIII. Specifically, K_1 gives up CNN layers to extract

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TABLE VIII

CLASSIFICATION ACCURACIES, INCLUDING OA (%), AA (%), PR (%), AND $F1$ ($\times 100$), ON THE THREE DATASETS TO PROVE THE NECESSITY OF NETWORK MODULES

Ablation Structure	MUUFL				Trento				Houston			
	OA (%)	AA (%)	PR (%)	F1 ($\times 100$)	OA (%)	AA (%)	PR (%)	F1 ($\times 100$)	OA (%)	AA (%)	PR (%)	F1 ($\times 100$)
K_1	88.92	71.27	84.35	74.01	97.86	97.86	97.49	97.62	87.45	82.35	84.86	83.35
K_2	89.29	71.71	82.39	75.12	98.53	97.81	98.14	97.95	87.94	84.64	84.35	84.33
K_3	88.86	71.91	83.70	74.90	98.52	97.91	98.18	98.01	88.12	82.74	85.25	83.75
K_4	89.76	70.98	81.49	73.58	98.48	97.48	98.61	98.02	88.76	84.19	84.99	84.39
K_5	89.46	70.47	81.89	73.51	98.65	98.17	98.29	98.22	88.81	84.03	86.12	84.63
K_6	89.15	68.73	80.92	71.05	98.39	96.73	97.94	97.31	88.07	83.93	84.59	84.11
K_7	89.15	71.67	83.21	74.87	98.27	97.44	97.84	97.59	88.96	84.29	84.26	83.99
K_8	87.35	68.14	77.49	70.71	98.31	96.18	98.21	97.14	84.26	73.95	82.23	76.02
K_9	89.29	68.81	86.38	71.70	98.10	96.94	97.61	97.25	88.33	84.18	85.70	84.77
K_{10}	89.58	73.47	84.71	75.10	98.75	97.89	**98.77**	98.32	88.32	82.55	86.92	84.36
K_{11}	88.89	68.71	82.05	72.62	98.26	97.26	97.66	97.40	88.87	84.60	86.19	85.25
K_{12}	88.69	69.56	83.22	72.65	98.55	97.54	98.46	97.99	88.58	84.11	85.65	84.66
Our	**90.55**	**74.01**	**87.61**	**76.68**	**98.99**	**98.56**	98.20	**98.38**	**89.41**	**84.78**	**86.95**	**85.75**

high-level features, meaning that the raw RS samples are delivered for the following modules. On the contrary, the accuracies of K_1 on the three datasets are lower than those of HugIpuNet, highlighting the essential role of the CPS module. In K_2 and K_3, the DSM features are not extracted and captured comprehensively. These structures cannot make full use of the potential of the DSMs. HugIpuNet improves the OA by 1.26%, 0.46%, and 1.47% over K_2 and by 1.69%, 0.47%, and 1.29% over K_3 on the three datasets. These accuracies demonstrate that neglecting this module tends to yield suboptimal performance. Furthermore, fixing $(\sigma + \sin(\beta t))^2$ in K_4 limits the fluctuation ability of the adjacent matrices and reduces the anti-interference ability. The results of the SDL-related structures prove that both SDL streams play crucial roles in feature extraction. Notably, HSIs generate superior classification performance by highlighting and enhancing spatial arrangements and spectrum distributions. The DLC module focuses on unifying illumination-wise properties and preserving discriminant information simultaneously. When removing this module in K_8, the effect of complex environmental scenarios on global–local information analysis cannot be fully limited, decreasing the OA by 3.2%, 0.68%, and 5.15% on these three datasets. K_9 and K_{10} preserve only one decomposed component, so that they lose some discriminant information and decrease classification performance inevitably. The results of K_{11} and K_{12} further verify that the detailed structures in the DLC module are significant and cannot be removed or replaced. Replacing the concatenation layer with the addition layer in K_{12} tends to form feature overlap and weakens the distinctiveness of latent features.

3) Necessity of Loss Elements: The loss function of HugIpuNet consists of four loss elements, including L_{Dec}, L_{Ref}, L_{Sha}, and L_C. To verify their necessities, the structures M_1–M_5 with multiple loss element couples are listed as:

Considering L_C is vital for classification, it cannot be removed in these loss element couples. The classification accuracies of these structures are presented in Table IX for comparison. Specifically, these losses play crucial roles in guiding the entire optimization process toward finding the optimal direction. The absence of any loss element will result

TABLE IX

LOSS ELEMENT COUPLE EXPLANATIONS AND CLASSIFICATION ACCURACIES, INCLUDING OA (%), AA (%), PR (%), AND $F1$ ($\times 100$), ON THE THREE DATASETS TO PROVE THE NECESSITY OF LOSS ELEMENTS

Ablation Structure	Loss				Metrics	Datasets		
	L_{Dec}	L_{Ref}	L_{Sha}	L_C		MUUFL	Trento	Houston
M_1	×	✓	✓	✓	OA	89.05	98.46	88.09
					AA	69.37	97.14	83.76
					PR	84.19	97.78	84.77
					F1	72.88	97.44	84.06
M_2	✓	×	✓	✓	OA	89.22	98.61	87.91
					AA	70.82	97.84	83.11
					PR	82.65	**98.71**	84.71
					F1	72.83	98.26	83.84
M_3	✓	✓	×	✓	OA	89.37	98.53	88.96
					AA	69.39	98.25	84.40
					PR	79.72	98.13	85.60
					F1	72.70	98.17	84.64
M_4	✓	×	×	✓	OA	89.07	98.62	88.83
					AA	69.54	97.92	83.83
					PR	85.54	98.20	86.12
					F1	73.14	98.05	84.77
M_5	×	×	×	✓	OA	88.86	96.78	86.12
					AA	71.14	95.12	79.71
					PR	81.94	96.39	82.37
					F1	73.66	95.74	80.75
Our	✓	✓	✓	✓	OA	**90.55**	**98.99**	**89.41**
					AA	**74.01**	**98.56**	**84.78**
					PR	**87.61**	98.20	**86.95**
					F1	**76.68**	**98.38**	**85.75**

in a decline in classification accuracy. These accuracies further emphasize that the decomposition-related losses (i.e., L_{Ref} and L_{Sha}) are significant in the whole network. The OAs of M_4 on three datasets are 1.48%, 0.37%, and 0.58% lower than those of HugIpuNet. Moreover, M_2 and M_3 limit the decomposition from distinguishing aspects. These results can verify that the performance of HugIpuNet is closely related to these loss elements. The information interaction ability between two heterogeneous RS modalities can be preserved by retaining all loss elements.

4) Sliding Window Side Selection: The optimal sliding window side was chosen within $\{3, 5, 7, 9, 11\}$. The classification accuracies and training times per epoch with different ws are presented in Table X. Generally, larger ws contribute to delivering more spatial–spectral elevation information for network optimization and smaller ws can limit the interference

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TABLE X

CLASSIFICATION ACCURACIES AND TRAINING TIMES PER EPOCH WITH DIFFERENT SLIDING WINDOW SIDES ws ON THE THREE DATASETS TO SELECT THE OPTIMAL w FOR HUGIPUNET

Datasets	Slide Windows	Metrics				
		OA (%)	AA (%)	PR (%)	F1 (×100)	Time (s)
MUUFL	3	88.39	70.53	84.99	72.35	**4.38**
	5	**90.55**	**74.01**	**87.61**	**76.68**	5.35
	7	89.50	70.45	81.43	73.37	14.31
	9	89.38	71.40	80.47	73.75	20.11
	11	87.83	65.67	79.61	67.95	27.41
Trento	3	97.11	96.11	96.87	96.48	**2.48**
	5	**98.99**	**98.56**	98.20	**98.38**	2.76
	7	98.60	97.00	98.06	97.44	4.11
	9	98.71	96.47	**98.23**	97.30	7.33
	11	97.94	96.58	96.83	96.60	8.74
Houston	3	86.26	80.24	83.70	81.55	**9.07**
	5	**89.41**	**84.78**	86.95	**85.75**	10.51
	7	88.43	84.49	87.10	85.55	15.06
	9	**89.41**	83.75	**87.70**	85.11	18.05
	11	89.33	80.20	84.29	80.55	23.66

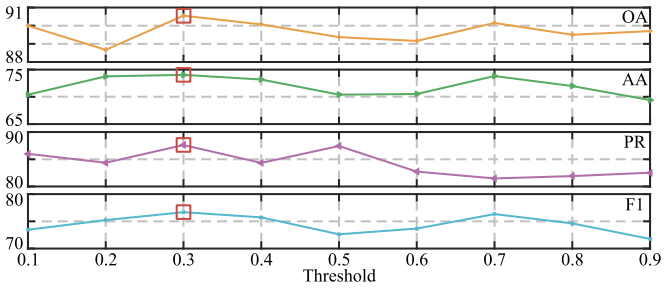


Fig. 13. Classification accuracies, including OA (%), AA (%), PR (%), and $F1$ (×100), on the MUUFL dataset to select the optimal threshold τ. The best accuracies are highlighted by red squares.

of irrelevant surrounding samples on feature extraction and improve the computational efficiency. Thus, selecting one satisfactory w is crucial for achieving superior performance. Apparently, the accuracies and the training time per epoch increase as the sliding window side increases. However, larger ws always result in more redundant structures, leading to suboptimal classification performance. When w grows from 5 to 7, the OA is decreased by 1.05%, 0.39%, and 0.98% while the time per epoch is increased by 8.96, 1.35, and 4.55 s on these three datasets, showing that larger ws always lead to lower efficiencies. When $w = 11$, the OA becomes 87.83%, 97.94%, and 89.33%, which are not satisfactory. Thus, $w = 5$ was chosen as the default sliding window side in this article.

E. Threshold Selection

In this article, the threshold τ in (3) was chosen within $\{0.1, 0.2, \ldots, 0.9\}$ on the three datasets since its value is related to the performance of the HUGCN layers. Generally, different thresholds reveal distinct node characteristics, allowing flexibility for the following optimization procedure. Using too large or too small τs cannot accurately describe node similarities in the HUGCN module. For a clear description, the classification accuracies on the MUUFL dataset with different τs are provided in Fig. 13. The results show that $\tau = 0.3$ is one satisfactory choice considering these four metrics comprehensively. The conclusions drawn from the three datasets are consistent. Thus, $\tau = 0.3$ was chosen as the default threshold.

V. CONCLUSION

In this article, a novel joint classification network called HugIpuNet has been proposed to strengthen the coupling capability between HSIs and DSMs and reduce the effect of illumination property variations in complex environmental scenarios. The core motivation of this work is to unify global illumination descriptions with assistance from elevations to capture precise multimodal properties specifically. The geological connection is considered to improve fusion interpretability and joint classification performance. The whole structure consists of four modules, including CPS, HUGCN, SDL, and DLC. Multiple illumination-wise mechanisms and HU graphs are considered to enhance the attribute unification ability in complex environmental scenarios. Furthermore, HugIpuNet performs well in analyzing global–local sample correlations. Experiments on three datasets prove that HugIpuNet outperforms other state-of-the-art techniques qualitatively and quantitatively. Multiple ablation and parameter selection experiments have also been conducted to fully reflect the effectiveness of HugIpuNet. Further experiments will focus on enhancing network robustness in various RS capturing scenarios, such as different weather conditions, complex terrains, and varying illumination levels.

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